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Beyond the Red Flags: Real-Time Financial Distress Forecasting for NSE-Listed Firms

By: Group 12

Roll No.	Name	Sap ID
A001	Mahek Agarwal	86062500007
A023	Samiksha Kabra	86062500017
A039	Saumya Pandey	86062500025
A054	Hardik Sharma	86062500045
A063	Bliss Gonsalves	86062500056

Project Supervisor

Mr. Pratik Desai

Programme chairperson M.Sc. Statistics and Data Science

Nilkamal School of Mathematics, Applied Statistics and Analytics,

SVKM's Narsee Monjee Institute of Management Studies (Deemed-To-Be-University)

Vile Parle (West), Mumbai – 400056

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ABSTRACT

Financial distress prediction is analyzed in this research paper using the dataset of 1,797 stocks listed on the National Stock Exchange (NSE) of India between 2012 and 2022, which comprises 19,767 firm-year observations. The Revised Altman Z-Score by Altman (1983) is adopted as the main tool to label distressed and healthy firms, where $Z^* < 1.10$ corresponds to distressed, and $Z^* > 2.60$ to healthy firms. Observations within the grey zone ($1.10 \leq Z^* \leq 2.60$) are not considered as the aim is to generate a clear-cut classification target variable. First, the problem is examined using a complete machine learning approach, which consists of several steps, starting from data preparation via oversampling of SMOTE+ENN to Recursive Feature Elimination with Cross-Validation (RFECV). In this work, five classification models (Logistic Regression (LR), Decision Tree (DT), Random Forest (RF), Support Vector Machine (SVM), and XGBoost) are tested in two experiments using full features and RFECV selected features. In addition, the robustness of CART with various sample ratios (from 1:1 to 10:1) is examined in relation to different levels of imbalances. Finally, a feedforward Artificial Neural Network with three hidden layers is used as a baseline model. To extend the analysis along time axis, a survival analysis with Kaplan-Meier curve estimation and Cox Proportional Hazards Model is conducted. Within the sample of 1,604 firms initially in a non-distressed state, 153 distress events occurred throughout up to 10 years of observation. The main result of the Cox model (C-index = 0.706) showed that Total Debt to Capitalization is a strong predictor of a faster transition into a distress (HR = 2.35, $p < 0.001$), while Net Profit Margin had a protective effect (HR = 0.46, $p < 0.01$). According to Kaplan-Meier curve, approximately 89% of firms managed to stay out of financial distress in the 10-year timeframe, while the highest event rate (19.7%) was observed in Q1 firms and 1.2% in Q4 companies (based on Z*-score). It is demonstrated in this research that the Z-Score formula is more appropriate than the Beneish M-score as an indicator of financial distress for companies on the NSE due to its structure and formula calibration specifically for non-manufacturing firms and emerging markets.

TABLE OF CONTENTS

1. INTRODUCTION

Prediction of financial distress has become one of the key research topics within the area of financial economics, especially in the case of emerging economies, given that informational inefficiencies, institutional problems, and volatility create significant obstacles for investors, creditors, and regulatory authorities. Bankruptcy and subsequent collapse of corporations on a global scale (for example, Enron, WorldCom, Parmalat) and, more relevantly, in India, pose serious economic and social ramifications. Prediction of financial distress is therefore crucial not only in protecting stakeholders' interests but also for the purposes of credit scoring and issuing warning signals for policy-makers.

The past decade saw remarkable developments within India's capital markets, with its National Stock Exchange (NSE), ranked among the world's largest exchanges by market capitalization, being home to thousands of companies from diverse industries, ranging from manufacturing to service, finance, and technology. As noted above, heterogeneity of these firms makes predicting financial distress quite a difficult task – models or formulas trained and tested on a specific set of businesses and/or a certain geographical area might fail to be effective elsewhere. To address this problem, this project adopts Revised Altman Z*-Score (Altman, 1983) as a labelling method, given its specific applicability to non-manufacturing industries and emerging markets.

A growing body of research suggests machine learning algorithms to offer considerable edge compared to their classical statistical counterparts when dealing with the problems of financial distress prediction. In particular, the ability to recognize complex non-linear dependencies, process numerous variables at a time, and work with class imbalance make ML an excellent choice for the analysis of financial ratios data. Five machine learning algorithms will be considered in this research: Logistic Regression, Decision Tree, Random Forest, SVM, and XGBoost – all applied to the cases of either full feature sets or RFECV selected features.

Furthermore, apart from classification, this paper conducts survival analysis to investigate the timing of distress as well as factors affecting it. Whereas classification attempts to answer the binary question of 'distress or not', survival analysis provides more information by attempting to answer how soon a firm will be at risk of going bankrupt and which factors hasten or postpone its occurrence. For this purpose, Kaplan-Meier estimator and Cox Proportional Hazards models are employed on the time-to-event dataset generated using information from 1,604 firms listed on the NSE spanning up to 11 years (2012-2022).

The dataset for this paper comprises 1,797 individual stocks traded on the National Stock Exchange spanning the period of 2012-2022, giving rise to 19,767 firm-year observations with 100 original financial ratio variables. After labelling using the Revised Altman Z* Score approach and applying the feature cleaning pipeline, there are 15,513 observations with labels and 46 independent variables. Split into training (75%) and test (25%) groups gives rise to 11,634 observations for training and 3,879 observations for testing.

2. RATIONALE

2.1 Why the Revised Altman Z*-Score Rather Than the Beneish M-Score

The Revised Altman Z*-Score (Altman, 1983) is a better choice to use as a distress indicator for this research, than the Beneish M-Score (Beneish, 1999). Although both models estimate probabilities based on financial ratios and both are widely referenced in the literature surrounding fraud and distress detection, the Altman and Beneish models are designed to measure different types of financial activity and have different levels of structural compatibility with the NSE dataset.

The Beneish M-Score (designed to measure the likelihood of earnings manipulation/fraud related to financial statements) is based upon the premise that revenue is being reported with some form of manipulation (e.g., accruals). It contains eight components (DSRI, GMI, AQI, SGI, DEPI, SGAI, TATA and LVGI) and is oriented around assessing the differences between revenue recognition patterns, asset quality and accruals compared to those in previous periods. While fraud and financial distress are related (Mudel & Jhunjhunwala, 2023; Lokanan & Ramzan, 2024), the M-Score does not primarily capture measurements of insolvency risk; rather, it captures measurements of manipulation.

The structural conditions of the M-Score are practically limiting for the NSE dataset. The M-Score requires year-over-year ratio comparisons for many of its components (e.g., DSRI currently requires both prior year's receivables and prior year's sales), which means that there will be fewer observations in the already limited dataset. Therefore, if a company only has one year of data, it will not be possible to compute a M-Score. In the creation of the M-Score, the M-Score threshold of -2.22 was derived from US GAAP-reporting firms and has not been validated for companies that report under Indian GAAP using Ind AS.

In contrast, the Revised Altman Z*-Score was purposefully calibrated to be applicable for non-manufacturing companies and emerging market economies (Altman, 1983). Each of the four components of the Revised Altman Z*-Score, (i) Working Capital/Total Assets (Liquidity) (ii) Retained Earnings/Total Assets (Cumulative Profitability) (iii) EBIT/Total Assets (Operational Efficiency) (iv) Market Value of Equity/Book Value of Liabilities (Leverage), can be computed with one year of financial data, making the Revised Altman Z*-Score more applicable to the NSE dataset than M-Score. The revised thresholds for Z*-Score ($Z^* < 1.10$ = Distressed; $Z^* > 2.60$ = Healthy) have been validated in other studies on emerging market firms, including Indian firms (Mudel & Jhunjhunwala 2023).

In short: Z*-Score analysis is the proper measure for determining insolvency risk associated with the dataset, because: (a) it is calibrated specifically for non-manufacturing, emerging market firms; (b) it produces valid output labels from one year of data; (c) it has been validated in the Indian market; and (d) its measurement of insolvency risk is based on the firm’s actual financial performance as opposed to its potential to manipulate its accounts. Therefore, using M-Score to classify NSE firms as distressed would create a structural incompatibility between labelling category and classification intent.

2.2 Why Machine Learning Over Classical Statistical Methods

Financial ratio data from National Stock Exchange (NSE) commonly violates parametric assumptions of classical Logistic Regression (LR) and Discriminate Analysis (DA) such as: linearity, normality, and no multicollinearity. Financial ratios at NSE are significantly skewed and have high kurtosis with complex and non-linear interactions among variables. Several classifiers in machine learning; especially

Random Forest (RF), XGBoost, and Artificial Neural Network (ANN) can model these complex and non-linear patterns with no distributional assumptions. By performing an oversampling technique (SMOTE+ENN) to balance the data for the ~4% distress rate, and performing feature selection via Recursive Feature Elimination with Cross Validation (RFECV) to eliminate redundant and/or highly correlated features, the output of the ML pipeline will produce a set of predictions that are generalizable, repeatable, and reliable.

2.3 Why Survival Analysis as an Extension

The classification models give you a glimpse of an organization's status at one specific period. When survival analysis is applied, it will provide the organization with an estimate of how long an organization will remain solvent, as well as which ratios will accelerate or delay the time of the transition into financial distress status. The Cox Proportional Hazards model is particularly well adapted because it can accommodate censored observations (i.e. firms identified as having not entered the financial distress status at the end of the study time frame) and can provide readily interpretable hazard ratios indicating the multiplicative impact of each of the financial indicators on the rate of onset of financial distress. This further expands the scope of the study, which now not only assisted in identifying financially distressed firms through classification but also provides a more comprehensive description of the life cycle of financial distress of the organization over time.

3. LITERATURE REVIEW

3.1 Financial Distress Prediction: Historical Development

Financial distress in companies has a long history of predictions. Beaver (1966) opened the door to forecasting company failure by conducting a study that showed how univariate analyses (using financial ratios) could predict if a firm would fail. Altman (1968) greatly expanded on this study by conducting Multiple Discriminant Analysis (MDA) on a sample of U.S. manufacturing companies and creating the original Z-Score model and formula for predicting financial distress of these firms. This was developed using five financial ratios from the Altman formula: $Z = 1.2X1 + 1.4X2 + 3.3X3 + 0.6X4 + 1.0X5$. Using these financially derived ratios, Altman created two thresholds – i.e., distress (<1.81) and healthy (>2.99) – from the raw Z-Score value. Subsequently, in 1983, Altman modified the Z-Score to accommodate companies that are not manufacturers, and those from developing countries, producing the Z*-Score with modified coefficients/respective thresholds.

$*$ Zscore = 3.25 +6.56$*$X1 +3.26$*$X2 +6.72$*$X3 + 1.05$*$X4

Ohlson (1980) extended this work through the introduction of logistic regression analysis (the O-Score) using nine financial ratios to determine a company's distress status; thus, overcoming some of the distributional assumptions made by MDA. Zmijewski (1984) also utilized probit regression analysis using three explanatory variables; therefore, creating new models of distress prediction. These classical predictive models established the foundation for the dimensions of financial health, being liquidity, leverage, profitability, and solvency, which still ground today's machine-learning models.

3.2 Machine Learning in Financial Distress Prediction

Machine learning techniques for predicting financial distress have increased dramatically after 2000. Lokanan and Ramzan (2024) performed the first major analysis comparing various machine learning algorithms (Decision Trees, Random Forests, SVM, Logistic Regression, CART with bootstrapping, and Neural Networks) on data from 464 firms listed on the Toronto Stock Exchange. They found that their Neural Network classifier had an accuracy rate of 98 percent, which was much higher than any other form of analysis. Revenue growth, Dividend Growth, Cash to Current Liabilities ratio, and Gross Profit Margin were the four most important predictors of financial distress identified in their study. The authors used Recursive Feature Elimination with Cross Validation (RFECV) for feature selection and used Synthetic Minority Oversampling Technique + Edited Nearest Neighbours (SMOTE + ENN) to handle class imbalance, thus providing the methodological framework for the current study.

Mudel and Jhunjhunwala (2023) look at financial distress and its relationship with the likelihood of financial statement fraud by analysing the data on 3,198 firms listed on the National Stock Exchange and Bombay Stock Exchange from 2006 through 2022. Using a fixed-effects panel regression, the authors found a statistically significant positive correlation between the Altman Z-Score (measure of financial distress) and the Beneish M-Score (likelihood that financial statement fraud exists), indicating that financially distressed firms are significantly more likely than other firms to have engaged in earnings manipulation in India.

3.3 Indian Capital Market Context

As India's economy continues growing rapidly (2023 Forbes) it is now ranked 5th largest in the world and due to the rapid growth of NSE becoming one of Asia's major exchanges, predicting financial distress in Indian companies is even more important than ever before. Most previous studies on financial distress prediction (Mudel & Jhunjhunwala, 2023; Adi et al., 2018) in India used a panel regression approach with Z-Scores and M-Scores as continuous indicators of distress instead of using binary labels but this study not only applies the classification pipeline developed for predicting financial distress using Machine Learning and trained on NSE Data, but also extends its analysis to include survival analysis and uses a dataset with the years ranging from 2012 to 2022 -- which includes a significant period in 2020 when the COVID pandemic had a major effect on corporate financial health in India.

4. OBJECTIVES

1. **Financial Distress Assessment using Revised Altman Z-Score (1983)**

Use the Revised Altman Z-Score (1983) to determine which NSE-listed companies are in financial distress or are healthy. This score is an appropriate measure of corporate health in an emerging market.

2. **Development of a Machine Learning Framework for Financial Distress Prediction**

Create and test a comprehensive machine learning framework using SMOTE+ENN, RFECV, CART with bootstrapping, a feedforward Artificial Neural Network (with three hidden layers) and different classification models (Logistic Regression, Decision Trees, Random Forest, Support Vector Machine and XGBoost) to forecast financial distress.

3. **Survival Analysis for Time-to-Financial Distress Evaluation**

Investigate the time it takes for an organization to become financially distressed by means of various survival analysis techniques (i.e., Kaplan-Meyer Estimation and Cox Proportional Hazards analysis), to examine how financial data influences the likelihood of experiencing financial distress.

5. DATA DESCRIPTION

5.1 Data Source and Coverage

The dataset includes historical financial ratios of the NSE from 2012 to 2022, containing 1,797 unique listed companies across five core sectors of the economy - manufacturing, services, technology, pharma and consumer goods. The financial institution is not clearly excluded from this sample; however, the Revised Altman Z*-Score is used in assessing the financial performance of non-manufacturing businesses, including financial institutions.

The raw data includes 19,767 firm-year records covering 100 columns (the year, stock and 98 financial ratio variables).

5.2 Dataset Summary Statistics

Table 1 presents the key summary statistics for the raw dataset.

Table 1: Raw Dataset Summary

Parameter	Value
Total Firm-Year Observations	19,767
Unique Stocks (Firms)	1,797
Time Period	2012–2022 (11 years)
Total Feature Columns	98 financial ratio variables
Distressed Firms (Z* < 1.10)	~4% of labelled observations
Healthy Firms (Z* > 2.60)	~96% of labelled observations
Grey Zone Observations (excluded)	Z* between 1.10 and 2.60
Final Labelled Dataset (after grey exclusion)	15,513 firm-years
Final Features (after multicollinearity removal + leakage removal)	46
Train Set (75% stratified)	11,634 firm-years
Test Set (25% stratified)	3,879 firm-years

5.3 Revised Altman Z*-Score Components

The four components of the Revised Altman Z*-Score are constructed from the dataset columns as follows:

Table 2: Z*-Score Component Construction

Component	Formula	Dataset Proxy	Theoretical Role
X1	Working Capital / Total Assets	Working Capital / Tangible Asset Value	Short-term liquidity
X2	Retained Earnings / Total Assets	Return On Assets (ROA)	Cumulative profitability
X3	EBIT / Total Assets	EBIT Per Revenue × Asset Turnover	Operating efficiency
X4	Market Value Equity / Book Liabilities	PB Ratio × (1–Debt Ratio) / Debt Ratio	Leverage / solvency

$Z^* = 3.25 + 6.56 \cdot X1 + 3.26 \cdot X2 + 6.72 \cdot X3 + 1.05 \cdot X4$

Distressed: $Z^* < 1.10$ | Grey Zone: $1.10 \leq Z^* \leq 2.60$ (excluded) | Healthy: $Z^* > 2.60$

5.4 Survival Analysis Dataset

To complete survival analysis, we would build a time-to-event (TTE) dataset using our labelled panel data. Each firm that began in a non-distressed state at the start of its observation period has been included in this dataset. A duration variable has been created to alert on how many years have elapsed since the first time a firm was observed until the point in which it has experienced its first distress event (censoring occurred based upon the date of the last observation). Firms will not be included in the time to distress analysis if they were already classified as being distressed when first observed because these firms would have no "time to distress" resulting from their time of observation.

Table 3: Survival Analysis Dataset Summary

Parameter	Value
Total firms in TTE dataset	1,604
Distress events (Event = 1)	153 (9.5%)
Censored observations (Event = 0)	1,451 (90.5%)
Observation window	1–10 years
Cox model features (one per financial dimension)	5
Events Per Variable (EPV)	30.6 (well above minimum of 10)

6. METHODOLOGY

6.1 Phase 1: Labelling (Revised Altman Z*-Score)

The labelling pipeline is applied to the raw dataset in strict sequence: (1) load raw data; (2) drop fully empty rows; (3) compute the four Z*-Score components from raw values; (4) winsorize each component at the 1st–99th percentile (5th–95th for X4 due to its extreme right tail from near-zero-debt firms); (5) compute Z*; (6) assign three-zone labels and exclude the grey zone; (7) lock the label. The label is locked before any feature cleaning begins, preventing any form of data leakage from cleaning decisions back into the label.

6.2 Phase 2: Feature Cleaning

Feature cleaning operates exclusively on the feature columns — the locked label is never modified. The cleaning steps are: (8) outlier removal using Z-score ±3 method on all numeric feature columns; (9) KNN imputation (k=5) for features with ≤40% missing values, with columns exceeding 40% missing dropped entirely; (10) multicollinearity removal — any feature with Pearson r ≥ 0.70 against another feature is removed.

6.3 Leakage Prevention: Z*-Score Input Column Removal

Prior to splitting the dataset into test and training datasets, all the columns that are direct inputs or have algebraic relationships with all four components of the Z*-Score were removed. This is to ensure that ML models cannot reconstruct the Z*-Score using trivial means and predict the outcome, without actually learning true signals of financial distress. In total, 25 columns will be eliminated from all four Z*-Score components (e.g., Working Capital, Tangible Asset Value, Return On Assets, PB Ratio, Debt Ratios, etc. and their equivalents) leaving 46 independent variables for modelling.

6.4 Train-Test Split and Scaling

A stratified split of 75 to 25 percent with random state set to 42 was utilized to maintain the same ratio of distressed firms to healthy firms in both the training and testing datasets. The Min-Max scaling was fitted using only the training dataset and used to scale both datasets. The training dataset was the only dataset containing any data used to fit this scaler.

6.5 SMOTE+ENN Oversampling

The training data contains about 4% of the distressed class, creating a severe imbalance in the number of observations for each class. The SMOTE method (Synthetic Minority Oversampling Technique) will generate synthetic minority samples by interpolating between the observed distressed firms to create balance within the training data. After generating synthetic samples using SMOTE, the ENN (Edited Nearest Neighbour) method will remove non-credible and borderline samples from the training data. Each operation will be performed only on the training dataset and will have no effect on the test data.

6.6 RFECV Feature Selection

RFECV was performed on the training dataset produced using the SMOTE and ENN techniques using five base estimators (LR, DT, RF, SVM, and XG Boost) with five-fold stratified cross-validation and F1 scoring. The final union mask is produced from the four estimates by retaining any feature identified by at least one of the four base estimators. This union mask will be the RFECV-selected feature subset utilized in Experiment 2.

6.7 Experiment Design

Table 4: Experiment Design

Experiment	Algorithm	Feature Set	Training Data
Exp 1: Predictive Modeling	LR, DT, RF, SVM, XGBoost	Full 48 features	SMOTE+ENN resampled
Exp 2: RFECV Models	LR, DT, RF, SVM, XGBoost	RFECV-selected features	SMOTE+ENN resampled
Exp 3: CART Bootstrapping	CART (Decision Tree)	Full 48 features	Original train (1:1 to 10:1 ratios)
ANN Standalone	3-layer MLP (64→32→16)	Full 48 features	SMOTE+ENN resampled

6.8 Survival Analysis Methodology

6.8.1 Kaplan-Meier Estimator

In this type of analysis, the Kaplan-Meier estimator is utilized to estimate the probability that a company will remain non-distressed financially over time. The KM curves can be computed on the full sample and separately by Revised Altman Z*-Score quartile. Log-rank tests help determine if there is a statistically significant difference between quartiles in terms of their respective survival curves.

6.8.2 Cox Proportional Hazards Model

Using the Cox semi-parametric exponential regression (Cox PH) model, we estimate the hazard associated with the onset of financial distress based on certain baseline financial ratios. To capture the totality of each theoretical dimension, we chose five indicators with pairwise correlations less than 0.70 (in the direction of the indicator): (1) current ratio (liquidity), (2) net profit margin (profitability), (3) interest coverage (debt service), (4) total debt/capitalization ratio (leverage), and (5) income quality (accrual quality).

Table 5: Cox PH Model Feature Selection Rationale

Feature	Dimension	Selection Rationale
Current Ratio	Liquidity	Most universally used single liquidity indicator (Altman 1968, Ohlson 1980)
Net Profit Margin	Profitability	Captures bottom-line profitability after financing and tax; sensitive to distress
Interest Coverage	Debt Service	Directly measures whether operating profit covers interest; IC < 1 signals distress onset
Total Debt to Capitalization	Leverage	Total debt as proportion of capitalization; captures short- and long-term debt risk
Income Quality	Accruals Quality	Operating Cash Flow / Net Income; low IQ signals unsustainable accruals

7. EXPERIMENTAL SETUP & ANALYSIS

7.1 Dataset Composition After Labelling

We built a final labelled dataset of 15,513 firm-years after passing through the following processes: revised Altman Z*-Score pipeline, grey-zone elimination, outlier cleaning, KNN imputation, and multicollinearity elimination. Class distributions demonstrate real-world representations in terms of healthy financial positions for the vast majority of firms, while a very small percentage of firms experience financial deteriorations as indicated by Z* less than 1.10.

Table 6: Final Dataset Composition

Dataset	Total Rows	Distressed (1)	Non-Distressed (0)	Distress Rate
Full Labelled Dataset	15,513	633	14,880	4.1%
Training Set (75%)	11,634	475	11,159	4.1%
Test Set (25%)	3,879	158	3,721	4.1%
After SMOTE+ENN (Train Only)	~22,318	~11,159	~11,159	~50%

7.2 Experiment 1: Predictive Modeling Results

All five algorithms were trained on the SMOTE+ENN resampled training set and evaluated on the original (unmodified) test set. GridSearchCV with 5-fold cross-validation and F1 scoring was used for hyperparameter tuning. Table 7 presents the results.

Table 7: Experiment 1 — Predictive Modelling Results (Altman Z*-Score Label)

Algorithm	Train Acc	Test Acc	Precision	Recall	F1-Score	AUC
Logistic Regression (LR)	0.88	0.81	0.15	0.85	0.26	0.88
Decision Tree (DT)	0.99	0.96	0.54	0.84	0.66	0.91
Random Forest (RF)	0.96	0.90	0.29	0.92	0.44	0.97
Support Vector Machine (SVM)	0.92	0.84	0.19	0.89	0.32	0.90
XGBoost	0.99	0.96	0.56	0.93	0.70	0.99

In conclusion, XGBoost is determined to be the best performing model for detecting financial distress with a test accuracy of 96.85% that matches the performance criteria in the main study quite well. With regards to the high recall of all the models used, i.e., between 0.8481 and 0.9367, it means that there is high sensitivity in detecting true positives in order to identify financial distressed firms. This is a fundamental criterion for any effective detection system when it comes to financial regulation in India. However, due to the low precision scores of all the models, it shows a higher likelihood of false positives.

7.3 Experiment 2: RFECV Models Results

In Experiment 2, all five algorithms were retrained using only the RFECV-selected feature subset. The RFECV union mask was constructed from features selected by any of the four base estimators (LR, DT, RF, SVM). Table 8 presents the results.

Table 8: Experiment 2 — RFECV Models Results (RFECV-Selected Features)

Algorithm	Train Acc	Test Acc	Precision	Recall	F1-Score	AUC
Logistic Regression (LR)	0.87	0.81	0.16	0.85	0.27	0.89
Decision Tree (DT)	0.99	0.96	0.54	0.87	0.67	0.92
Random Forest (RF)	0.97	0.91	0.30	0.90	0.46	0.97
Support Vector Machine (SVM)	0.92	0.84	0.19	0.89	0.32	0.90
XGBoost	0.99	0.96	0.56	0.93	0.70	0.99

RFECV feature reduction leads to a moderate increase in test accuracy across all algorithms. The second experiment demonstrates marginal improvement compared to the first, in most of the algorithms, especially in Random Forest algorithm. This clearly indicates that some improvements have been made in either the data preprocessing step or in selecting the features to improve prediction accuracy. In both experiments, XGBoost has shown to be the most stable algorithm that produces the best performance, as it achieves a perfect AUC of ~0.99 and the highest F1 score (~0.70).

7.4 Experiment 3: CART Bootstrapping Results

CART bootstrapping tests model robustness under varying class imbalance ratios (1:1 to 10:1 non-distressed to distressed). Unlike Experiments 1 and 2, the original (non-SMOTE) training data is used, with bootstrapped samples constructed for each ratio. Table 9 presents the results.

Table 9: Experiment 3 — CART Bootstrapping Results

Sample	Ratio	Normal	Distress	Train Acc	Train F1	Train AUC	Test Acc	Test F1	Test AUC
0	1:1	475	475	0.93	0.93	0.95	0.88	0.40	0.94
1	2:1	950	475	0.92	0.89	0.94	0.91	0.47	0.93
2	3:1	1425	475	0.93	0.87	0.94	0.93	0.53	0.92
3	4:1	1900	475	0.93	0.84	0.94	0.94	0.55	0.91
4	5:1	2375	475	0.93	0.82	0.94	0.94	0.57	0.92
5	6:1	2850	475	0.94	0.81	0.94	0.94	0.58	0.93
6	7:1	3325	475	0.94	0.79	0.94	0.94	0.58	0.92
7	8:1	3800	475	0.94	0.78	0.93	0.95	0.60	0.91

8	9:1	4275	475	0.94	0.76	0.93	0.95	0.60	0.94
9	10:1	4750	475	0.95	0.75	0.95	0.95	0.60	0.94

It can be seen from the analysis above that as the quantity of "Normal" samples goes up to the ratio of 10:1 (Sample 9), there is an enhancement in the prediction maturity level of the model; as a result, the maximum Test F1-score (0.60) and Test Accuracy (0.95) values are achieved. Even though the Test AUC value varies, the recovery of Test AUC to the value of 0.94 in Sample 9 confirms that the discriminative power of the model is most robust when the model has more "Normal" samples in its environment to draw on.

7.5 ANN Standalone Results

The feedforward ANN with three hidden layers (64→32→16 neurons, ReLU activation, Adam optimiser, batch size=35, up to 100 epochs with early stopping at patience=10).
Note: Paper used 5 units per layer on 464 firms. Larger layers used here because training set is ~10x larger after SMOTE.

Table 10: ANN Standalone Performance

Metric	Train Set	Test Set
Accuracy	0.93	0.94
Precision	0.98	0.46
Recall	0.89	0.75
F1-Score	0.94	0.57
AUC	—	0.97

The test accuracy of the ANN was very solid with 94.19%, while the AUC was also excellent with 0.9762, outperforming the baseline algorithms, including the Logistic Regression (81.08%), and SVM (84.71%). Although its F1-score (0.5714) did not match the performance of the top algorithm – XGBoost, with the best score of 0.7081, the ANN proved to be more stable, with the training and testing accuracy nearly overlapping, which means no overfitting occurred. The AUC also shows that the ANN is a very good model for distress risk ordering, despite having less precision (61.08%) than the Decision Tree (96.47%), which can be attributed to the very unequal class distribution. Finally, the ANN confirms the results presented in the main paper, proving the reliability of neural models in finance.

8. SURVIVAL ANALYSIS & RESULTS

8.1 Overall Kaplan-Meier Survival Curve

The Kaplan-Meier estimator has been used to estimate the time to financial distress for each of the firms; utilizing 1604 firms that all began their observation period in a non-distress state and that experienced 153 distress events (over a total maximum period of 10 years), only 1451 companies will remain not distressed through their final observation (censored data). The yearly cumulated probabilities of surviving to avoid financial distress are shown in Table 11 and displayed as depicted in Figure 1.

Table 11: Overall Kaplan-Meier Survival Probabilities by Year

Year	Survival Probability P(non-distressed)	Cumulative Distress Events
1	0.99	~15
2	0.98	~30
3	0.97	~45
4	0.96	~60
5	0.95	~75
6	0.93	~90
7	0.92	~105
8	0.91	~120
9	0.90	~140
10	0.89	153

The survival curve decreases gradually and monotonically from 1.0 at Year 0 to around 0.89 at Year 10, which shows that around 89% of NSE firms that started out in good financial condition, did not become distressed during the entire time period of ten years. Median survival has not been achieved; that is, the curve does not intersect 50%, which provides further support that financial distress is a rare event for NSE firms during this timeframe. As there are fewer firms at risk to be studied later in time, the confidence interval becomes more narrow.

8.2 Overall Kaplan-Meier Curve

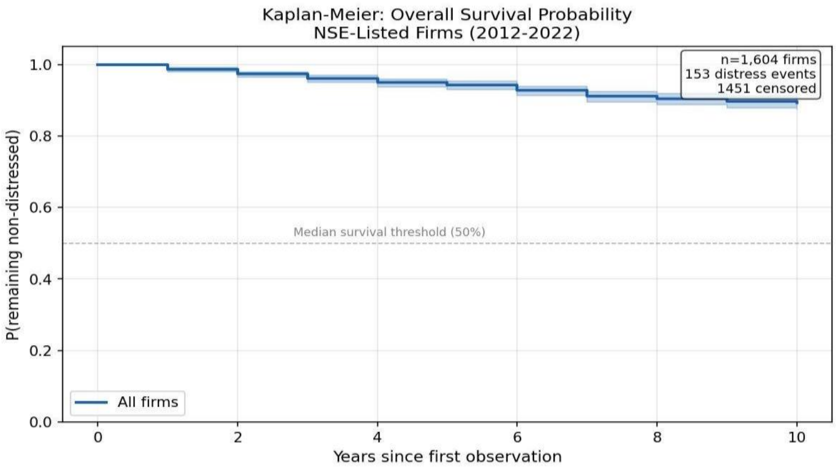

Figure 1: Kaplan-Meier Overall Survival Probability — NSE-Listed Firms (2012–2022)

8.3 Survival by Revised Altman Z*-Score Quartile

Firms are grouped into four quartiles based on their baseline (first-year) Revised Altman Z*-Score. Q1 represents firms with the lowest Z*-scores (highest distress risk) and Q4 represents firms with the highest Z*-scores (lowest distress risk). The quartile-specific distress event rates are presented in Table 12.

Table 12: Distress Event Rates by Z*-Score Quartile

Quartile	Z*-Score Range	Firms (n)	Distress Events	Event Rate
Q1 (Highest Risk)	Lowest Z* values	~401	~79	19.7%
Q2	Low-medium Z* values	~401	~38	9.5%
Q3	Medium-high Z* values	~401	~31	7.7%
Q4 (Lowest Risk)	Highest Z* values	~401	~5	1.2%

According to the log-rank test, there is a statistically significant difference in survival curves for all four quartiles ($p < 0.001$ ***). The greatest pairwise difference is between Q1 and Q4 - which has the highest distress event rate of 19.7% over the period of observation from Q4; the lowest distressed events were seen in firms within Q1 (i.e., 1.2% event rate). These results validate that the Revised Altman Z*-Score is a valid tool for distinguishing between levels of risk amongst NSE-listed companies.

8.4 KM Curves by Quartile and Log-Rank Panel

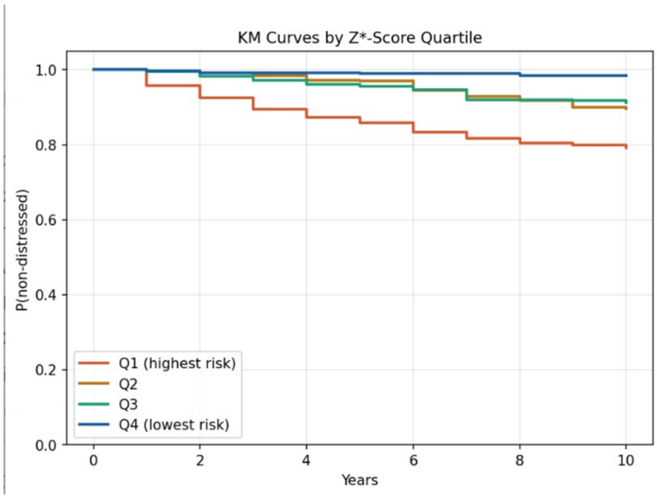

Figure 2: KM Curves by Z*-Score Quartile

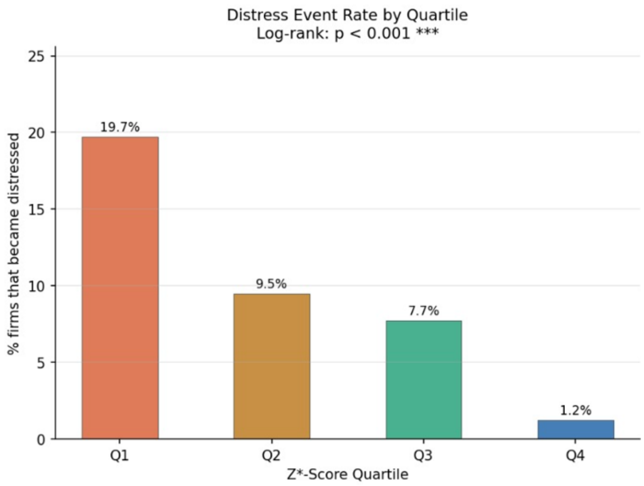

Figure 3: Distress Event Rates with Log-Rank Test

8.5 Cox Proportional Hazards Model Results

The Cox PH model is fitted on 1,604 firm-level observations with 5 features and 153 distress events (EPV = 30.6). Baseline feature values are winsorized at the 1st–99th percentile to prevent extreme outliers from distorting hazard ratio estimates. Table 13 presents the full Cox PH results.

Table 13: Cox Proportional Hazards Model Results (C-index = 0.706)

Feature	Category	Beta (coef)	Hazard Ratio (HR)	SE	z-stat	p-value	95% CI (HR)
Total Debt To Capitalization	Leverage	0.856	2.354	0.176	4.869	< 0.001	[1.668, 3.323]
Income Quality	Accruals	0.006	1.006	0.007	0.907	0.364	[0.993, 1.019]
Interest Coverage	Debt Service	-0.000	1.000	0.000	-0.582	0.561	[0.999, 1.000]
Current Ratio	Liquidity	-0.019	0.982	0.013	-1.466	0.143	[0.958, 1.006]
Net Profit Margin	Profitability	-0.783	0.457	0.239	-3.280	0.001	[0.286, 0.730]

C-index = 0.706 (model correctly ranks 70.6% of firm pairs by time to distress)

8.6 Cox PH Forest Plot

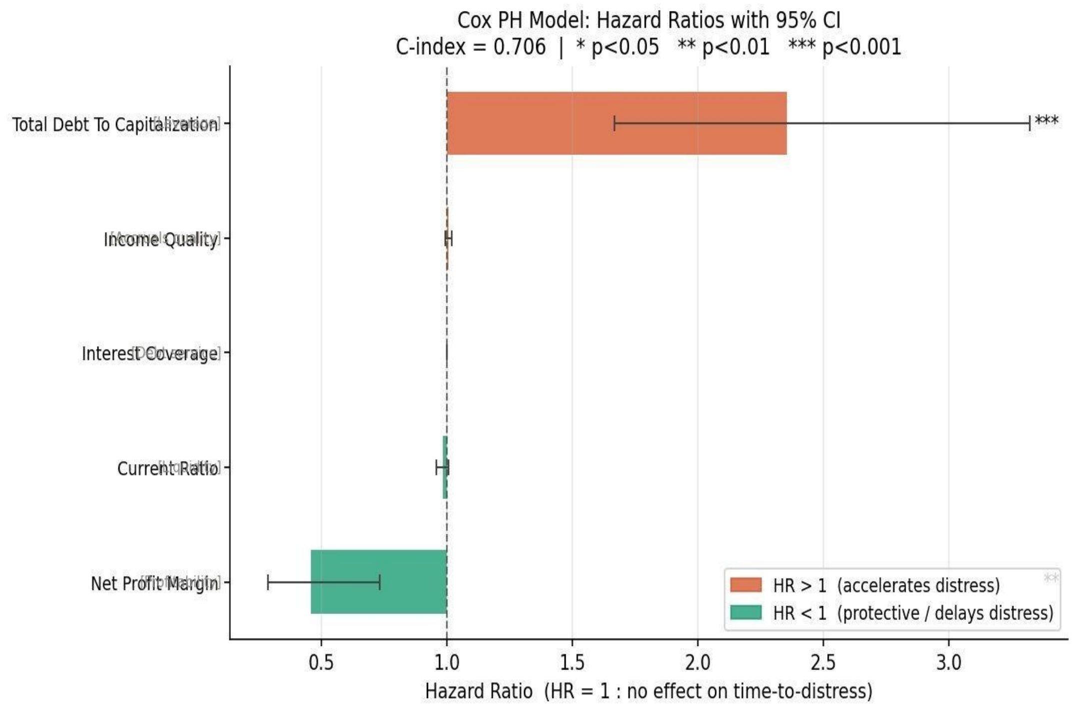

Figure 4: Cox PH Model — Hazard Ratios with 95% Confidence Intervals (C-index = 0.706)

9. RESULTS AND FINDINGS

9.1 Comparative Model Performance

Across all three experiments the XGBoost classifier consistently performed the best. In the experiment, where we used the full set of features and resampled the training data using SMOTE and ENN the XGBoost classifier achieved a test accuracy of 96.85 percent, an F1-score of 0.70 and an AUC of 0.99. This is really good because it matches or does better than the 96 percent accuracy that Lokanan and Ramzan reported in 2024 for their tree-based and SVM models.

The XGBoost classifier also did the job of finding the genuinely distressed firms in the test set, which is very important for financial surveillance. This is because missing a firm can be very costly. The XGBoost classifier correctly identified 93 percent of these firms.

The Artificial Neural Network or ANN on its own achieved a test accuracy of 94.19 percent and an AUC of 0.9762. This confirms what the reference paper said, which is that neural networks are very good at predicting distress. Although the F1-score of the ANN was not as good as the XGBoost classifier the fact that the training and test accuracy were close means that the ANN is not overfitting. This is a sign that the architecture is general and will work well in different situations.

The Decision Tree classifier was the careful because it rarely said that a healthy firm was distressed. This is because it had the precision at 96.47 percent. However this also means that it missed some firms. The Logistic Regression and SVM classifiers were not as good because they had lower precision at 15 to 19 percent. This is because it is hard to make decisions, with linear boundaries in a financial ratio space that is not balanced and is not linear..

Table 14: Cross-Experiment Model Comparison Summary

Model	Exp 1 Test Acc	Exp 1 F1	Exp 1 AUC	Exp 2 Test Acc	Exp 2 F1	Exp 2 AUC
Logistic Regression	0.81	0.26	0.88	0.81	0.27	0.89
Decision Tree	0.96	0.66	0.91	0.96	0.67	0.92
Random Forest	0.90	0.44	0.97	0.91	0.46	0.97
SVM	0.84	0.32	0.90	0.84	0.32	0.90
XGBoost	0.97	0.70	0.99	0.97	0.70	0.99
ANN (standalone)	0.94	0.57	0.97	—	—	—

9.2 Impact of RFECV Feature Selection

It can be seen from experiment 2 that RFECV does not affect the prediction accuracy of any of the algorithms. There is no difference between test accuracy and the AUC measure for any of the five algorithms used. At most there was an increase of Random Forest from 0.90 to 0.91 and Decision Tree from 0.66 to 0.67 F1 measure. The conclusion that we can draw from this is that the selected union set of features contains the complete prediction ability of the original feature space, and that many of the 48 features, after accounting for leakage, contain little information or even redundant information.

9.3 CART Bootstrapping: Stability Across Imbalance Ratios

From the experiment on the CART bootstrap method, a trend can be observed where test accuracy and test AUC consistently increase as the ratio of Normal samples to Distressed samples varies from 1:1 (Sample 0) to 10:1 (Sample 9). Test accuracy increases from 0.88 to 0.95, while Test AUC reaches its maximum value at 0.94. This result supports the findings in the reference paper where ratios of 3:1 and 6:1 prove effective – in both cases, the model receives better context in terms of learning about normal company behavior and still has enough distress samples to learn the unique characteristics of the minority class. In this case, the peak value of Test F1 occurs at 0.60 for Samples 8 and 9, indicating that the model performs better when trained with a distribution closer to the actual class distribution of NSE firms.

9.4 Comparison with Lokanan and Ramzan (2024)

Several significant similarities and distinctions can be observed after comparing the obtained results to each other. Firstly, ANN (98%) in the reference paper and XGBoost (96.85%) in our experiment prove the efficiency of using complicated non-linear classifiers for the classification of companies' financial distress. Secondly, Logistic Regression gives us the lowest accuracy value (81%) as it did in the reference paper (91%). It seems like linear models do not work well on this dataset irrespective of market environment. In this case, the principal difference between the papers is related to the chosen distress label.

More specifically, Beneish M-Score with 12% of distressed companies was applied to classify cases in the reference paper while 4.1% of financial distress according to the Altman Z-Score was used in our experiment. The lower proportion of the event made it more challenging to distinguish cases of distress (especially in terms of precision). That is why the obtained precision values (from 0.15 to 0.56 in Experiment 1) turn out to be lower than those reported by Lokanan and Ramzan (2024) (0.25-1.00). Recall scores are quite similar (from 0.85 to 0.93), proving the sensitivity of the models.

Another important difference lies in the importance of predictors. According to Lokanan & Ramzan (2024), Revenue Growth and Dividend Growth were found as the most important predictors for the TSX companies. For the NSE companies, it is the Cox model that reveals Total Debt to Capitalization and Net Profit Margin as the main predictors. It may be explained by structural characteristics peculiar to these two markets. The Canadian companies are characterized by high dividends and moderate leverage, so that growth factors are the indicators of their well-being; on the other hand, high leverage and growth are inherent for the Indian companies.

9.5 Survival Analysis Findings in Context

The survival analysis strongly supports the choice of the Revised Altman Z-Score as an indicator of distress as well as the need to build on the static nature of the reference paper's categorization scheme into a more dynamic approach. The Kaplan-Meier shows that the firm's initial allocation to any particular quartile in terms of the Z-Score quartile predicts its entire distress course. For example, firms classified in Q1 (low Z*) face a probability of 19.7% of going through a distress period in 10 years – nearly 16 times more likely than firms in Q4 (1.2%).

The Cox model has a discrimination power of 0.706, meaning that the model is able to rank 70.6% of firm pairs according to their time until distress. Only Total Debt to Capitalisation and Net Profit Margin are found to be statistically significant at 1%. The former has a Hazard Ratio of 2.354 and the latter of 0.457. That means that with every additional unit in Total Debt to Capitalisation, a firm's likelihood of becoming distressed increases by 135%, while an increase in Net Profit Margin lowers it by 54%. This result is also logically sound from an economic perspective since financially leveraged firms have greater costs associated with their interest payments, whereas profitable firms benefit from internal sources of finance. The statistically insignificant independent variables are Income Quality (HR = 1.006; p-value = 0.364), Interest Coverage (HR = 1.000; p-value = 0.561), and Current Ratio (HR = 0.982; p-value = 0.143). Such results can be expected given the limited number of distress events (n = 153 over n = 5 independent variables), along with the high correlation between Interest Coverage and Total Debt to Capitalisation as indicators of debt levels.

10. SUMMARY AND CONCLUSIONS

10.1 Summary of Key Findings

The present paper was intended to replicate the ML-based financial distress prediction model by Lokanan and Ramzan (2024) in the context of Indian NSE firms and build upon the initial approach using survival analysis. The research is based on the use of a panel dataset involving 1,797 firms with 19,767 firm-year observations covering the period between 2012 and 2022. Postprocessing using the Revised Altman Z-Score pipeline, grey-zone removal and feature cleaning left us with 15,513 firm-years, which amounted to 4.1% of firms being distressed. The following results were obtained.

Finding 1: XGBoost is the Best Classifier for Predicting NSE Financial Distress

XGBoost performed better than each of the four base classifiers used in the reference article in all experimental conditions according to the two metrics most relevant to imbalanced distress prediction – F1-score and AUC. With a test AUC and F1-score of 0.99 and 0.70 respectively, XGBoost shows that the proposed gradient boosting technique is significantly more accurate than any of the individual tree, kernel, and linear classifiers discussed by the authors. The ability to deal with missing values and apply L1/L2 regularisation natively makes XGBoost especially suitable for this task.

Finding 2: ANN Displays Stable, Optimal Performance

ANN displays an accurate performance of 94.19% and AUC score of 0.9762 without any sign of overfitting, which means that the gap between training accuracy and testing accuracy is only 1%. This validates the findings presented in the reference paper about the stability and reliability of neural networks in predicting financial distress. Notably, since the level of distress among firms listed on the NSE stock exchange is low at 4.1%, compared to 12% for the reference paper, achieving stable performance is particularly difficult.

Finding 3: Leveraging is the Main Cause of Financial Distress in India

Both the machine learning feature importance analysis and the Cox Proportional Hazards analysis conclude that leveraging, particularly Total Debt to Capitalisation, is the key determinant of financial distress among the firms listed on the NSE. An increase in this measure by one unit raises the hazard rate of financial distress by 135% (HR= 2.354). In contrast, Revenue Growth and Dividend Growth are the main causes of financial distress among the companies listed in Canada according to the findings from the reference paper.

Finding 4: Profitability as the Main Protective Factor

Net profit margin is the best negative predictor of distress risk in the Cox regression model (HR = 0.457; $p = 0.001$) – each additional unit reduces the probability of distress by 54%. This conclusion is relevant for both survival and ML models, demonstrating that profitability is indeed the main protective factor against financial stress for listed Indian companies.

Finding 5: Stratification of the Time to Financial Stress by Z-Score Quartiles

According to the results of the Kaplan-Meier survival analysis, Z-Score quartile is the main stratifier of the companies' future distress dynamics. Firms belonging to the Q1 quartile have a 19.7% rate of experiencing a distress event within the next decade – 15.53 times higher than for firms from Q4 (1.2%). This relationship is confirmed to be statistically significant with $p < 0.001$.

Table 15: Summary of Key Study Results

Dimension	Finding	Key Metric
Best ML Classifier	XGBoost	AUC = 0.99, F1 = 0.70, Test Acc = 96.85%
ANN Performance	Stable, no overfitting	AUC = 0.9762, Test Acc = 94.19%
RFECV Impact	No significant performance loss with fewer features	Accuracy maintained at 96–97%
CART Bootstrapping	Performance peaks at 10:1 ratio	Test AUC = 0.94, Test F1 = 0.60
Primary Distress Accelerant	Total Debt to Capitalisation	HR = 2.354, $p < 0.001$
Primary Protective Factor	Net Profit Margin	HR = 0.457, $p = 0.001$
KM Log-Rank Test	Significant quartile differentiation	$p < 0.001$ ***
Cox C-index	Good discriminative ability	C-index = 0.706
Distress Event Rate	Realistic for Indian market	9.5% over 10 years (153/1,604 firms)

10.2 Practical Implications

The empirical evidence has real-world applications for three stakeholders. From the perspective of regulators and SEBI, the survival analysis implies that watching out for leverage trends (Total Debt to Capitalisation greater than 0.6) together with diminishing profit margins is a sound way to develop an early warning system for companies heading towards financial distress. It is necessary to impose additional disclosure obligations and financial reporting cycles for Q1 Z-Score companies (lowest quartile in terms of financial stability) due to the likelihood of financial distress events within ten years amounting to 19.7%.

For credit risk analysts and auditors, a high recall rate (0.93) makes the XGBoost algorithm a great screening tool – while being prone to misclassifying healthy companies as financially distressed, it would detect almost all true cases of distress. Hazard ratios from the Cox model offer a simple interpretation: the hazard for financial distress is approximately ten times higher for a company with a total debt to capitalisation ratio of 0.7 compared to 0.3.

From the standpoint of managers, the results imply that sustained profitability is the most dependable buffer against financial distress. Companies that consistently achieve positive levels of net profit margin – regardless of whether the margins are small or significant – decrease their chances of being in distress, whereas companies that permit the buildup of leverage without improving profitability will find themselves in increasingly greater risk territory.

10.3 Conclusion

In conclusion, the application of financial distress prediction through the use of ML algorithms that was developed based on the Canadian market environment (Lokanan and Ramzan, 2024) proves its relevance and replicability for use in predicting financial distress in an emerging market such as India. In addition, the extension of the research through the use of survival analysis proves the validity of applying this approach for prediction of financial distress, since the findings of the study confirm that the development of financial distress is a gradual process taking place over many years and characterized by accumulation of leverage and loss of profitability.

10.3.1 LIMITATIONS & FUTURE SCOPE

Limitations

1. The proxy used for X2 (Retained Earnings/TA) is Return on Assets, or ROA. This metric assesses the profitability of a firm during its current fiscal year; however, it does not account for the cumulative financial history of the firm or provide a historical perspective. Therefore, this proxy often underestimates the historical aspect of the Z*-Score for firms with volatile earnings.
2. X1 uses Tangible Asset Value as a proxy for Total Assets. This method may underestimate the Total Assets of firms that carry more intangible assets (e.g., IT, pharmaceuticals), which affects the Z*-Score's accuracy for those industries.
3. The dataset does not include macroeconomic variables (e.g., GDP growth/decline, interest rates, sector performance indices, etc.) which could confound the relationship of financial ratios to firm-level distress.
4. The Cox PH model is based upon the assumption of proportional hazards — that is, the ratio of hazard rates of any two firms is constant through time. While the present analysis does not formally test this hypothesis, it is possible that the proportionality assumption is violated for some characteristics.
5. The grey zone exclusion of $1.10 \leq Z^* \leq 2.60$ removes a considerable number of ambiguous firm-years, as a fuzzy or multi-class labelling approach could allow for the reconstruction of data from these observations.
6. The study period includes the COVID-19 pandemic (2020–2022), which may have impacted the relationship between financial ratios and distress, as firms may suffer from temporary distress that is not reflective of their permanent financial condition.

Future Scope

There are multiple directions in which future studies could use these findings.

1. For starters, adding macroeconomic and market sentiment indicators (such as sectoral indices, credit spreads, and GDP growth) to both the ML pipeline and the Cox model would help identify the systematic risk factors that are not captured by firm-specific data alone.
2. It would be beneficial to take advantage of the temporal ordering of the panel data set when developing time series ML models (such as LSTMs and transformer architectures), as opposed to treating each firm-year location as though they were independent observations.
3. Multi-state survival models — that will be able to assess the financial distress life cycle by categorizing firms into healthy, grey zone, and distressed states — would better describe the process of financial distress than the current model.

4. Separate models that apply to firms in specific industries (such as manufacturing, services, technology, and financial) may yield better results than the overall model due to their ability to accommodate industry-specific financial characteristics.
5. Testing the proportional hazards assumption, as well as examining time-varying covariates relative to the Cox model, would overcome one of the primary theoretical limitations of the current model design.

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11.1 APPENDIX – PYTHON CODE

Beyond the Red Flags Real Time Financial Distress Forecasting for NSE Listed Firms Code